Contact_Angle_by_Hand: An Image Analysis Tool

Analyze fluid interfaces with precision and simplicity.

Overview: This software generates canny-edged images and provides pixel-to-real-world measurement conversions via linear regression. It was used to analyze images of fluid interfaces in low-gravity conditions.

Key Features

-Perform image analysis using a simple UI.

-Generate canny-edged images for deeper image analysis, highlighting significant edges to facilitate precise fluid interface detection.

-Click to draw Lines and points.

-Scroll to zoom in on the image for precise point notations.

-A slider to overlay and transition between the original image and the canny-edged image

-Line calibration is used to derive a unit line pixel-mm relationship, allowing users to convert measurements from pixel units to physical dimensions.

This is crucial for ensuring accurate quantitative analysis of fluid interfaces.

-Drag and drop precise lines.

-Automatic save and export images.

-Automatic save and export of line data to Excel sheets.

System Requirement

Required:

- **Operating System**: Windows 10 or later

- **Python Version**: Python 3.11 or later

- **Dependencies**: Listed in 'requirements.txt'

Additional Notes:

- `tkinter` is built into standard Windows Python installations.

- This software has not been tested on non-Windows platforms.

Installation

1. Clone the repository from GitHub using bash:

git clone https://github.com/casuzu/Contact-Angle-by-Hand.git

2. Navigate to the project directory:

cd Contact-Angle-by-Hand

3. Install the required dependencies:

py -m pip install -r requirements.txt

4. Run the application:

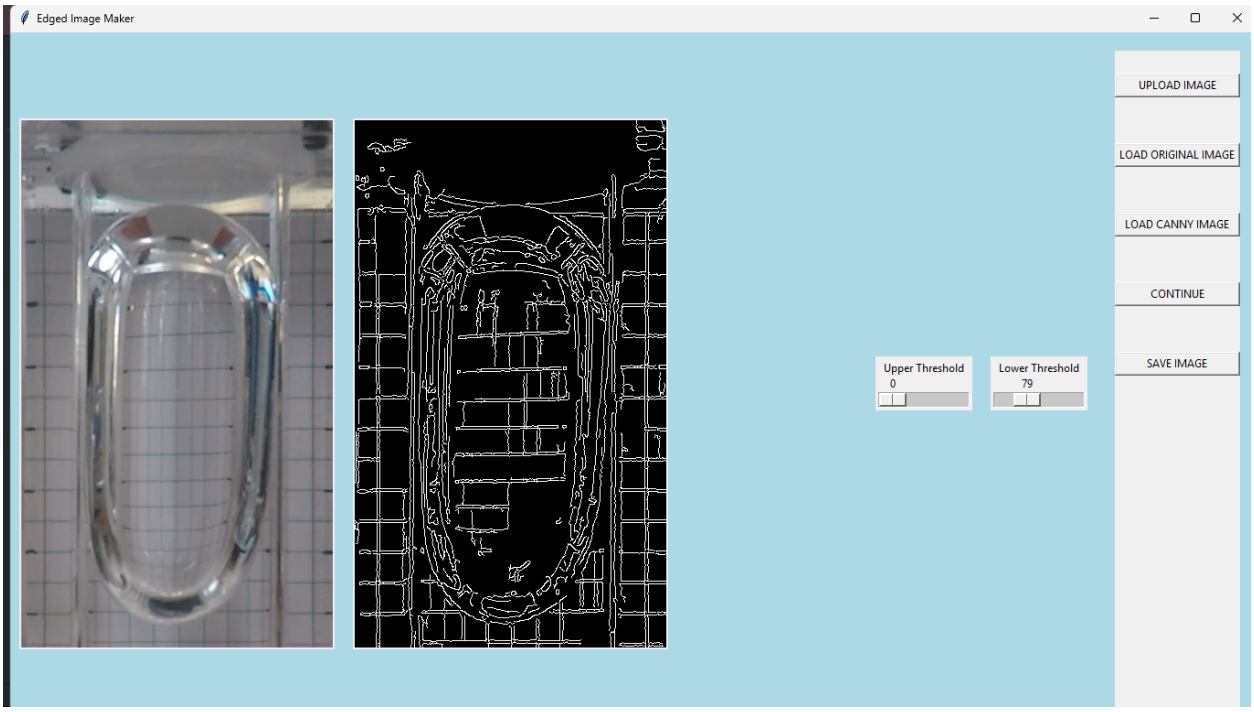
python main.py

##Usage

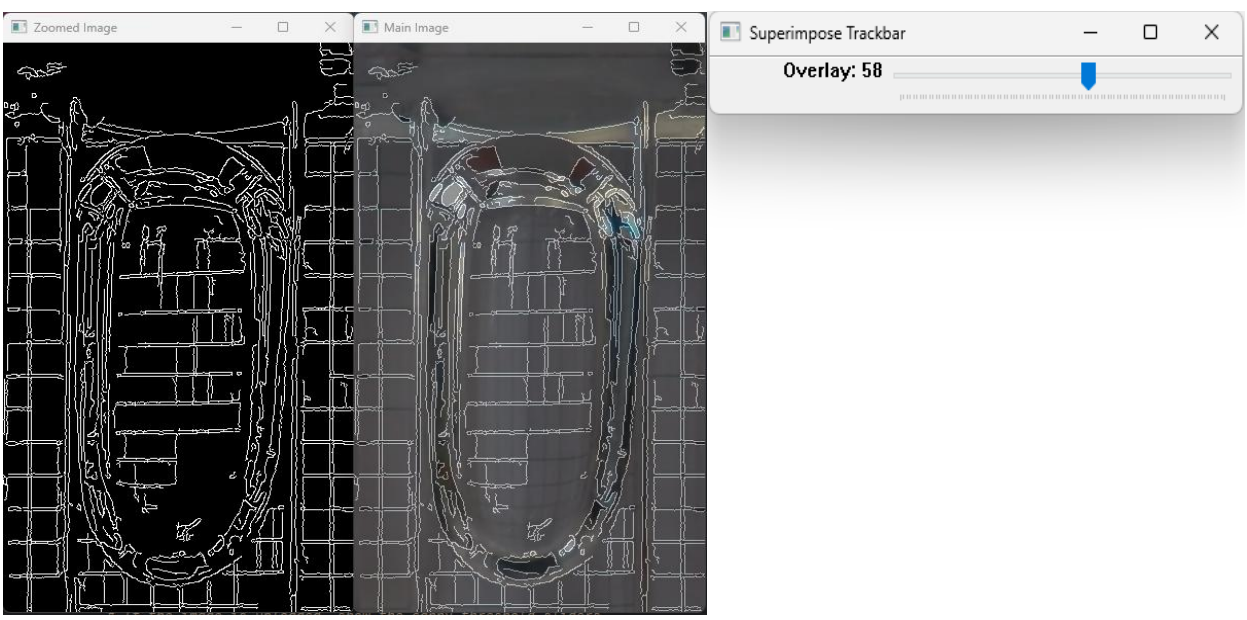
1. When using this application for the first time, use the “UPLOAD IMAGE” button to upload the image from which you want to generate canny-edge images. The image uploaded should preferably be in portrait orientation. Suppose the canny image was already created before. In that case, the “LOAD CANNY IMAGE” and “LOAD ORIGINAL IMAGE” buttons can be used to select the canny image saved and its original image and then select the “CONTINUE” button to proceed.

2. If the “UPLOAD IMAGE” button is selected instead, the original image(image on the left) can be converted to the canny image(image on the right) using the upper and lower threshold sliders. These sliders both go from 0 to 255 and are used to determine what constitutes an edge in the original image. After selecting the desired threshold values, the canny image can be saved to any location using the

“SAVE IMAGE” button. Selecting the “SAVE IMAGE” button will proceed to the analysis tool with selected original and canny images. UPLOADING THE IMAGE AND SELECTING THE “CONTINUE” BUTTON WITHOUT SAVING THE IMAGE WILL NOT LOAD THE ORIGINAL AND CANNY IMAGES TO THE ANALYSIS TOOL.



3. In the Image analysis tool, the superimpose trackbar can be used to slide between the canny and the original image. This change can be seen in the picture on the right below. The mouse scroll can zoom into the image on the right. The left image exists to display the area that was zoomed into.



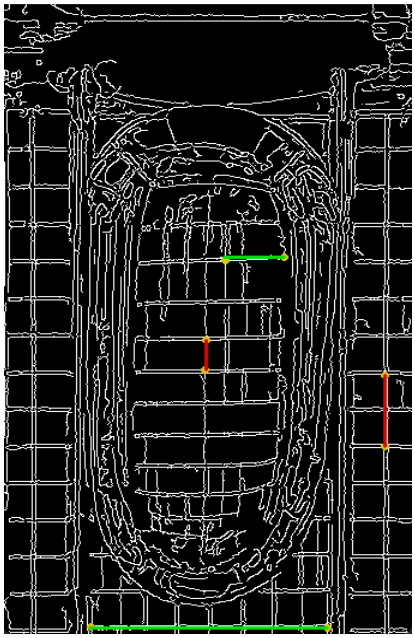
4. The main menu displays options for two modes in the terminal. Pressing the “L” key on the keyboard selects the “straight line” mode. The straight-line mode allows the user to choose any two points on the image, get the angle between them, adjust them to be a straight line, and then display the current distance and the former angle between these points. The previous and current coordinates of these points are also shown. This line is drawn on the right image as a red line if it is a vertical line and a green line if otherwise. Several lines can be drawn as needed. The mouse can also pick up any line and drag it to another coordinate on the image, and its coordinates are adjusted to reflect this change.

```
What mode would you like to run?:
Press l for Straight Line Mode.
Press c for Calibration Mode.
Press q to close the application.
Press m to see this menu again.
```

5. Pressing the “C” key on the keyboard selects the “calibration” mode.

```
========================
CALIBRATION MODE
========================
Enter 4 entries to calibrate the pixel to mm
 relationship.
Press the 'm' key after 'CALIBRATION COMPLETED'
 message to return to the main menu.
```

This mode generates a mathematical relationship between the line measurements calculated in pixels and the real-world measurements via linear regression. Making a line in this mode serves as an entry. Using the zoom function in this mode is advised for increased precision. The real-world line measurement for this line can then be entered in millimeters when prompted in the terminal window. Repeat this step three more times with variating line measurements to improve the accuracy of the linear relationship. The relationship is generated and displayed on the terminal when finished.



```
========================
CALIBRATION MODE
========================
Enter 4 entries to calibrate the pixel to mm
 relationship.
Press the 'm' key after 'CALIBRATION COMPLETED'
 message to return to the main menu.
Line distance in Pixels:  117.0
How many millimeters(mm) is this line?: 110
Line distance in Pixels:  74.0
How many millimeters(mm) is this line?: 73
Line distance in Pixels:  91.0
How many millimeters(mm) is this line?: 87
Line distance in Pixels:  36.0
How many millimeters(mm) is this line?: 29
========================
Estimated Relationship: Y = -4.7845 + 1.0004x
Estimated Accuracy: 99.29%
========================
CALIBRATION COMPLETED.
Press the 'm' key to return to the main menu.
```

6. It is advised to calibrate through the calibration mode before using the straight-line mode. After calibration, the line mode will display the pixel and real-world measurement in millimeters.

```
========================
STRAIGHT LINE
========================
Line Dist(pixels) =  72.00006944441095
Line Dist(mm) =  67.24681896887397
Angle =  89.92042257962264
 Previous Starting: [169, 234], Ending: [169.1, 306]
Adjusted Starting: [169, 234], Ending: [169, 306]
```

7. Pressing the “Q” key on the keyboard after any time will quit the application and produce two Excel sheet files in the application’s file location. These files will contain the vertical and horizontal line coordinates drawn in the straight-line mode. These results will also be displayed on the window terminal.

```
Lines = [[[169, 234], [169, 306]]]

Vertical Lines = [[[169, 234], [169, 306]]]

Horizontal Lines = []

Process finished with exit code 0
```

	A	B
1	[X1, Y1]	[X2, Y2]
2	[169, 234]	[169, 306]
3		
4		
5		